Proyecto EDA – Notas de exámenes

Hector Yustos

Hipótesis planteadas

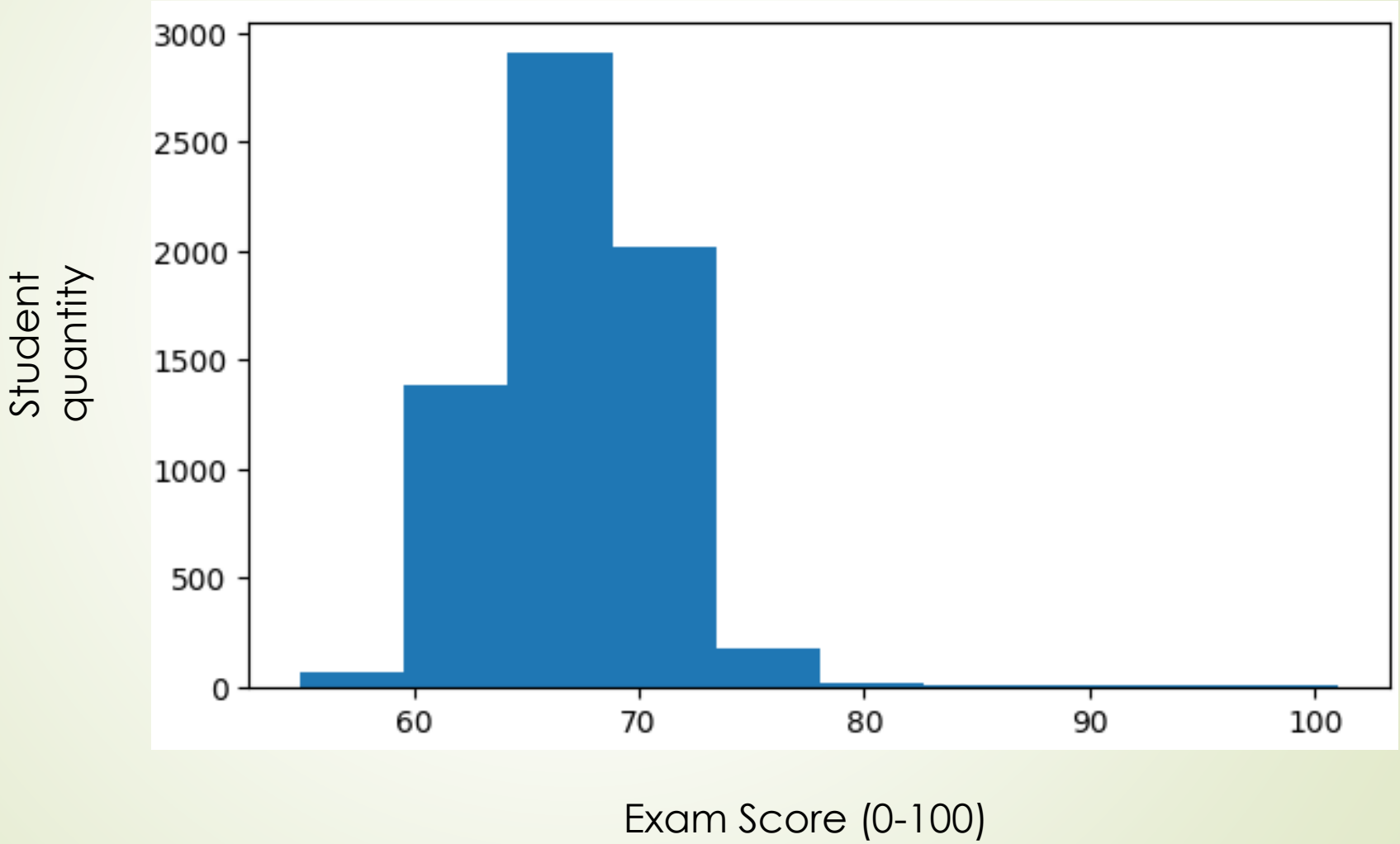
- Principal:
  - "Los estudiantes con una buena organización de estudio tienen más probabilidades de sacar mejor nota"
- Subhipótesis:
  - "La importancia de la motivación de un estudiante a la hora de la nota final"

Consideraciones

- Fuente: https://www.kaggle.com/datasets/grandmaster07/student-exam-performance-dataset-analysis
- Manteniendo la privacidad del estudio y los estudiantes, estos son los datos que se pueden revelar:
  - Prestigiosa universidad
  - Estudiantes de una asignatura en particular → 6.607 alumnos
  - Seguimiento durante todo el cuatrimestre de sus hábitos de estudio → 20 variables
  - Evaluación mediante el examen final de la asignatura (variable principal)

Nota Mínima	Nota Máxima
55	100

Consideraciones



Hipótesis principal

- "Los estudiantes con una buena organización de estudio tienen más probabilidades de sacar mejor nota"

Hours Studied

Parental Involvement

Access to Resources

Attendance

Internet Access

Teacher Quality

Distance from Home

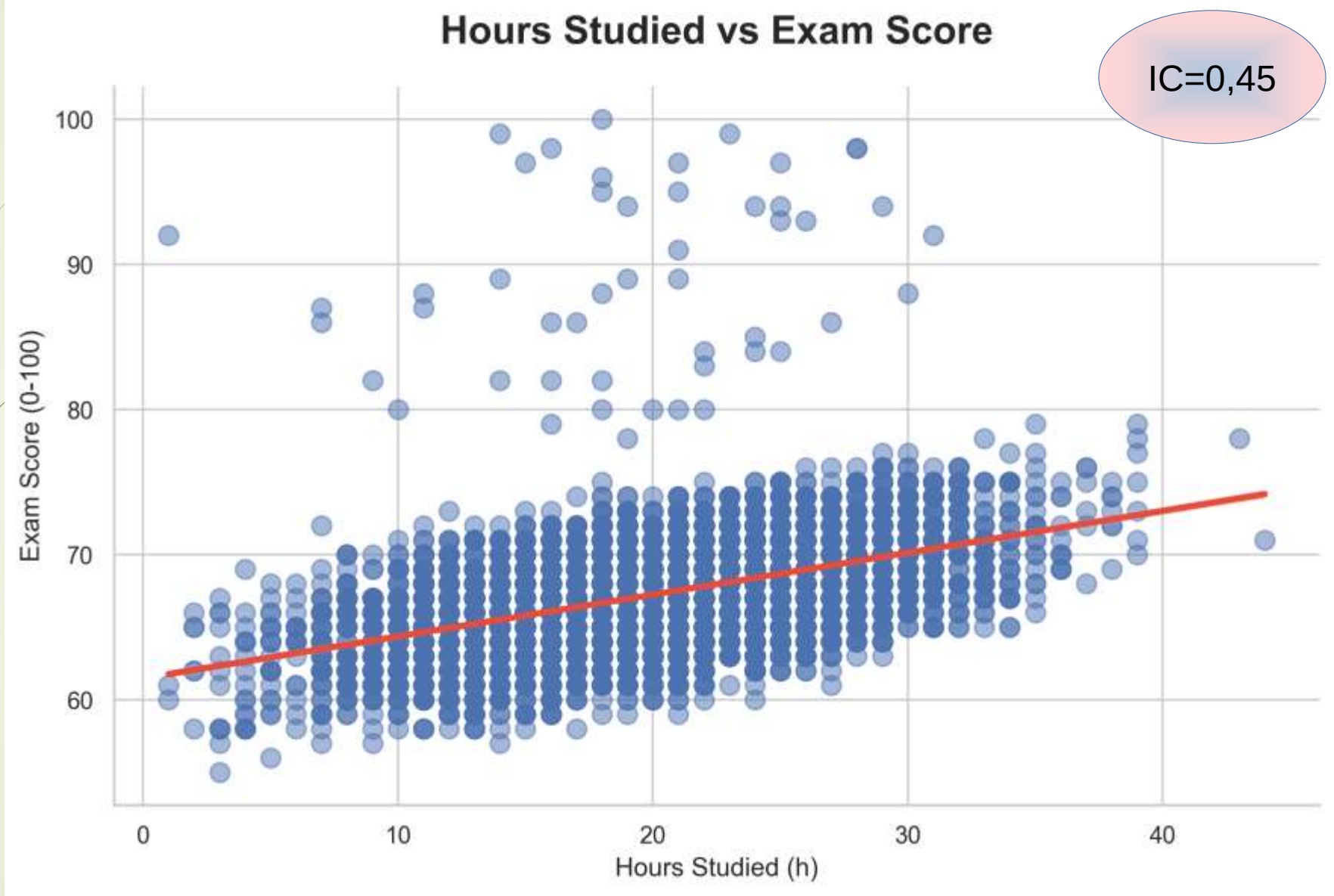
Sleep Hours

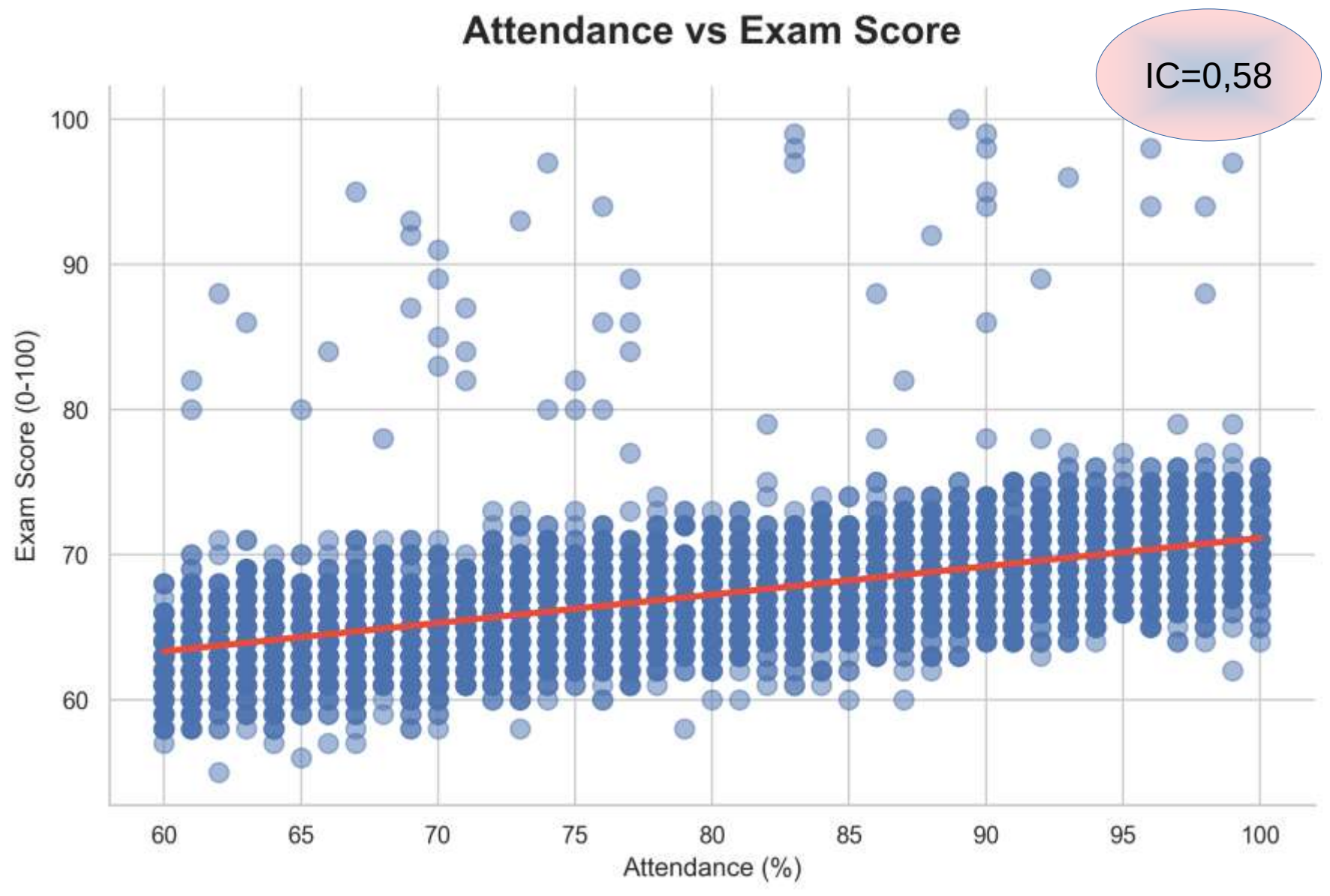
Physical Activity

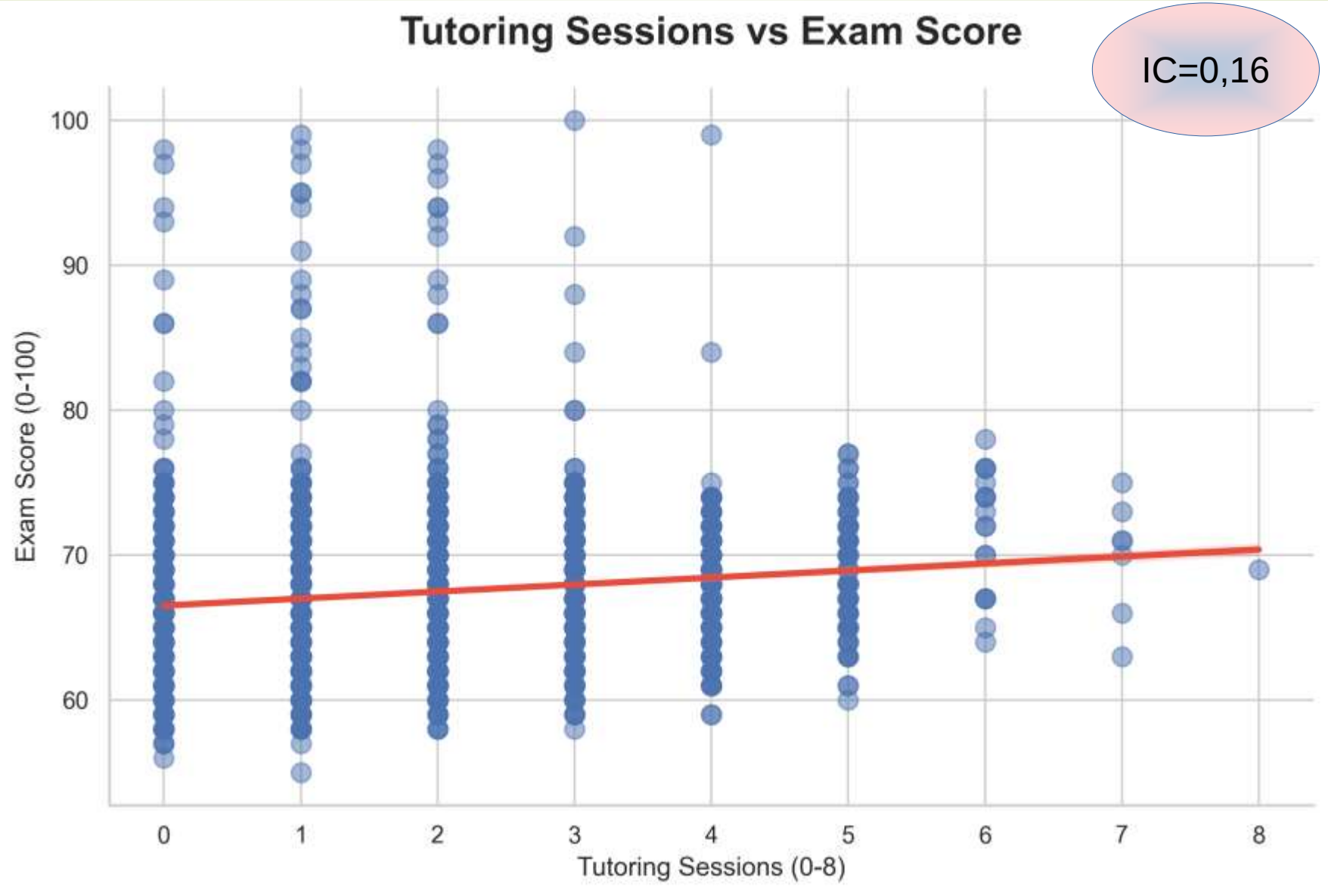
Tutoring Sessions

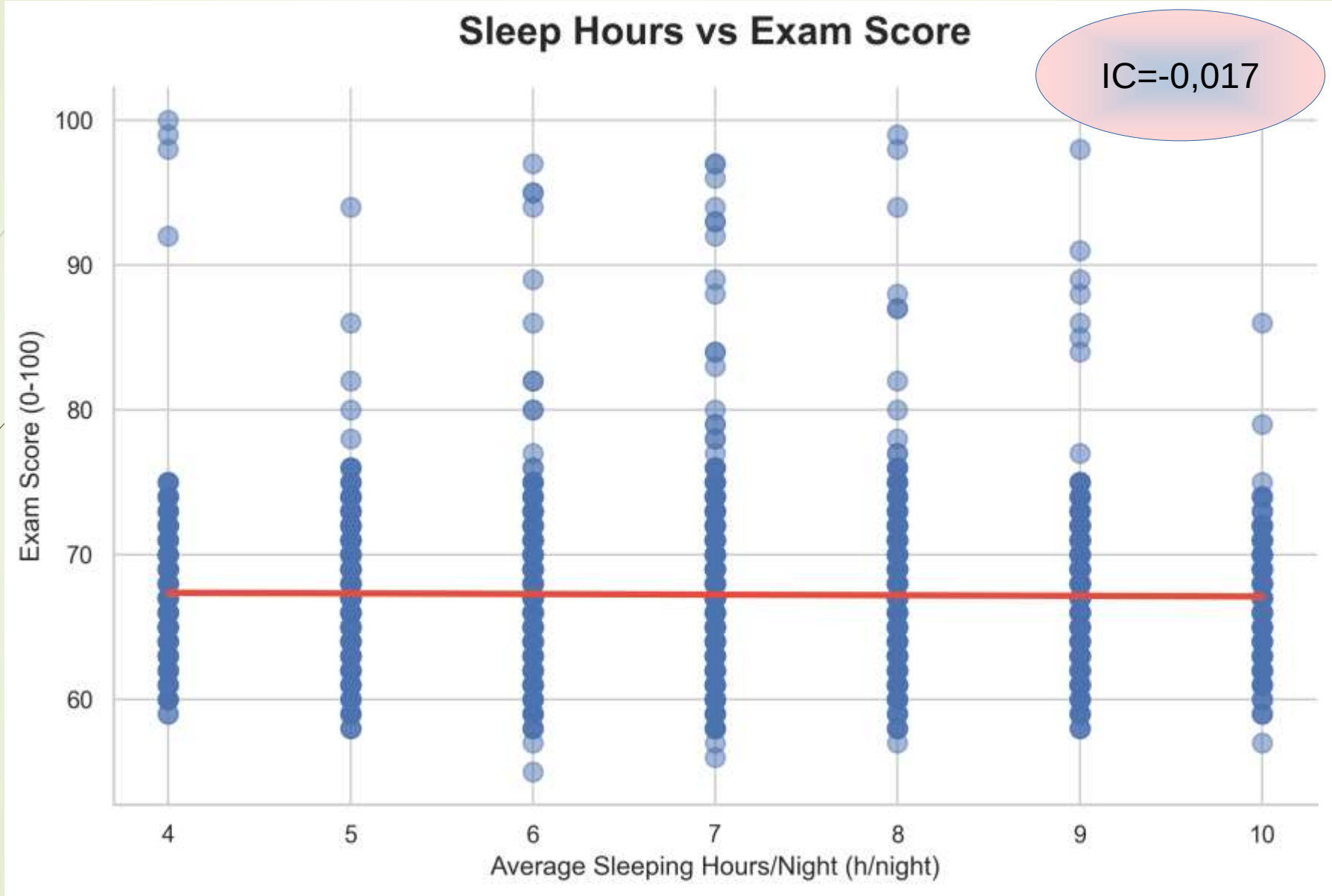
Learning Disabilities

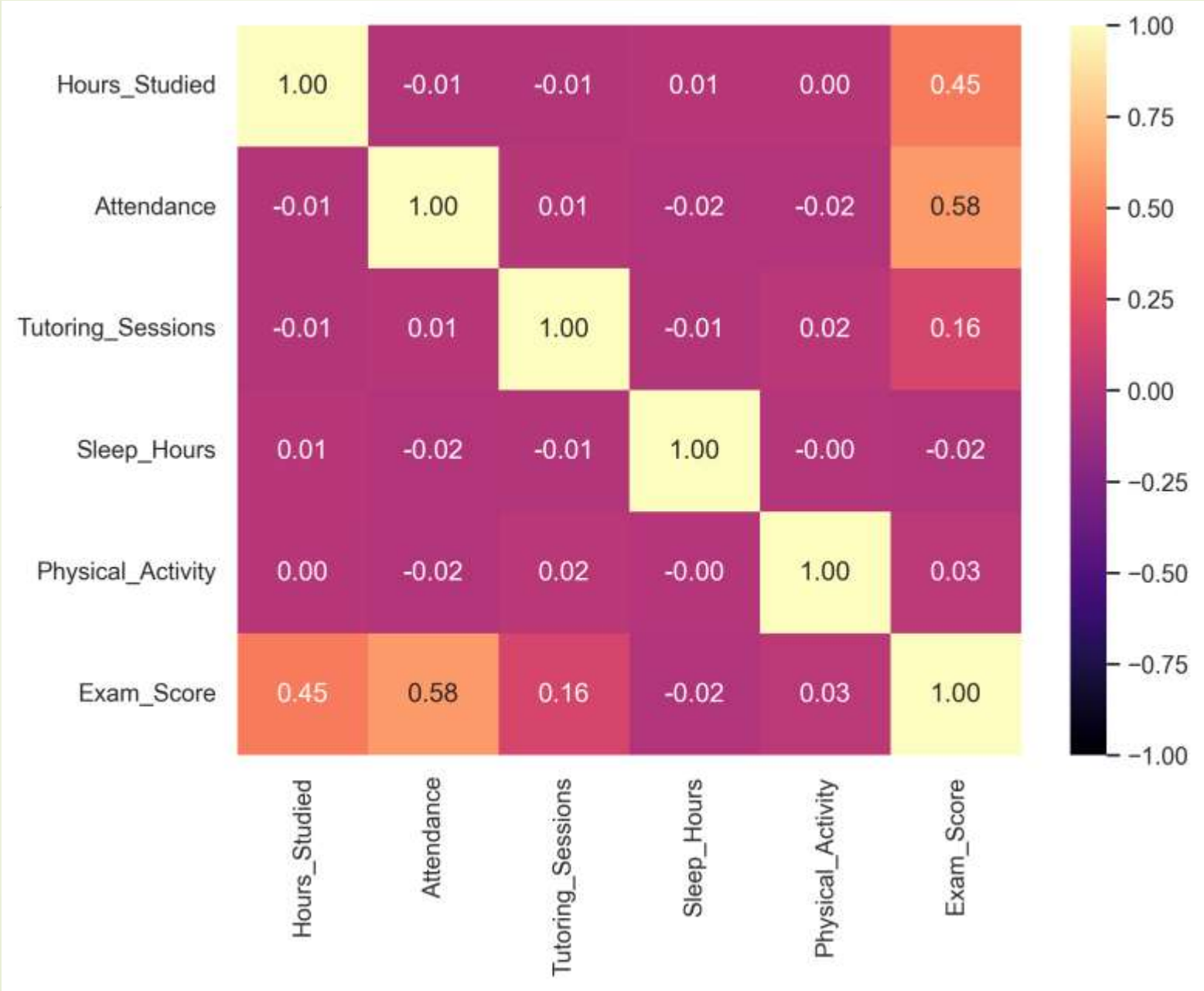
Extracurricular Activities


Hours Studied vs Exam Score
IC=0,45
Exam Score (0-100)
100
90
80
70
60
0
10
20
30
40
Hours Studied (h)

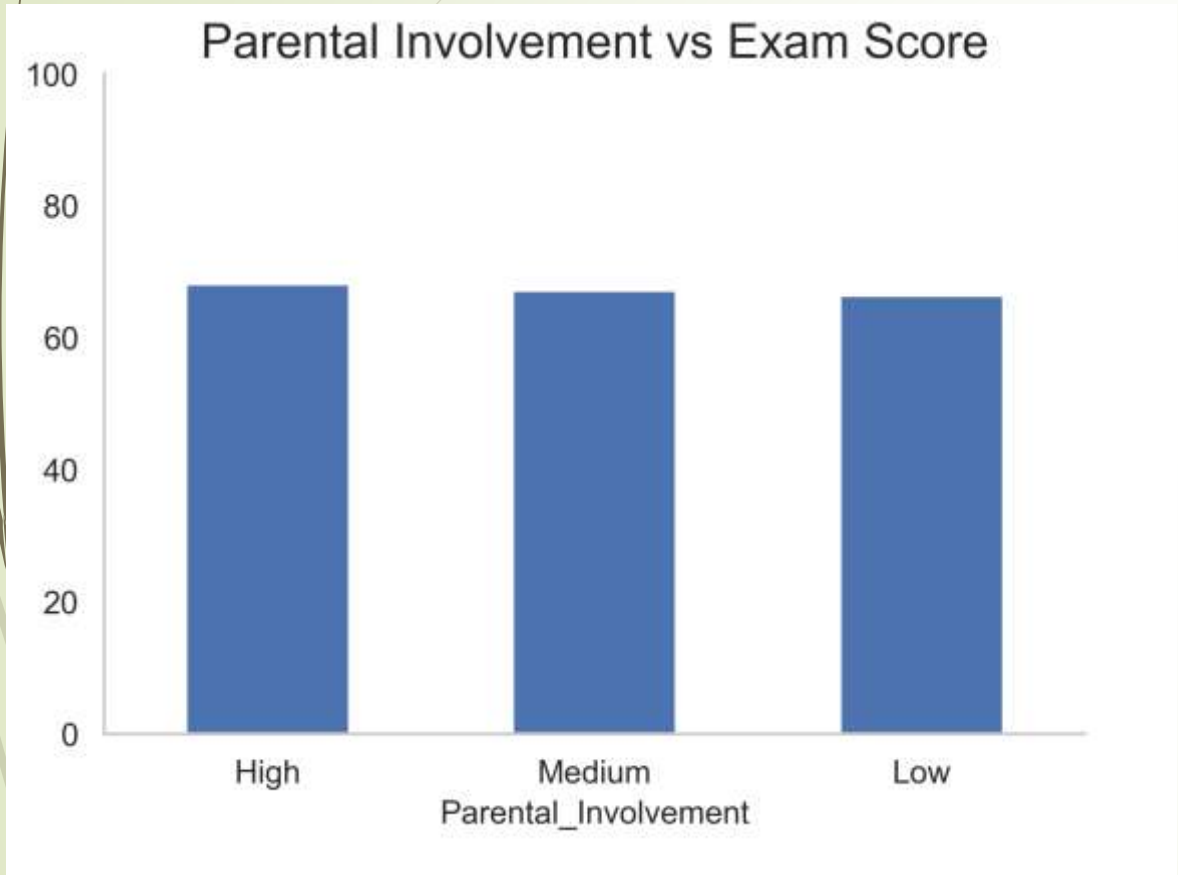

Attendance vs Exam Score
IC=0,58
Exam Score (0-100)
100
90
80
70
60
60
65
70
75
80
85
90
95
100
Attendance (%)

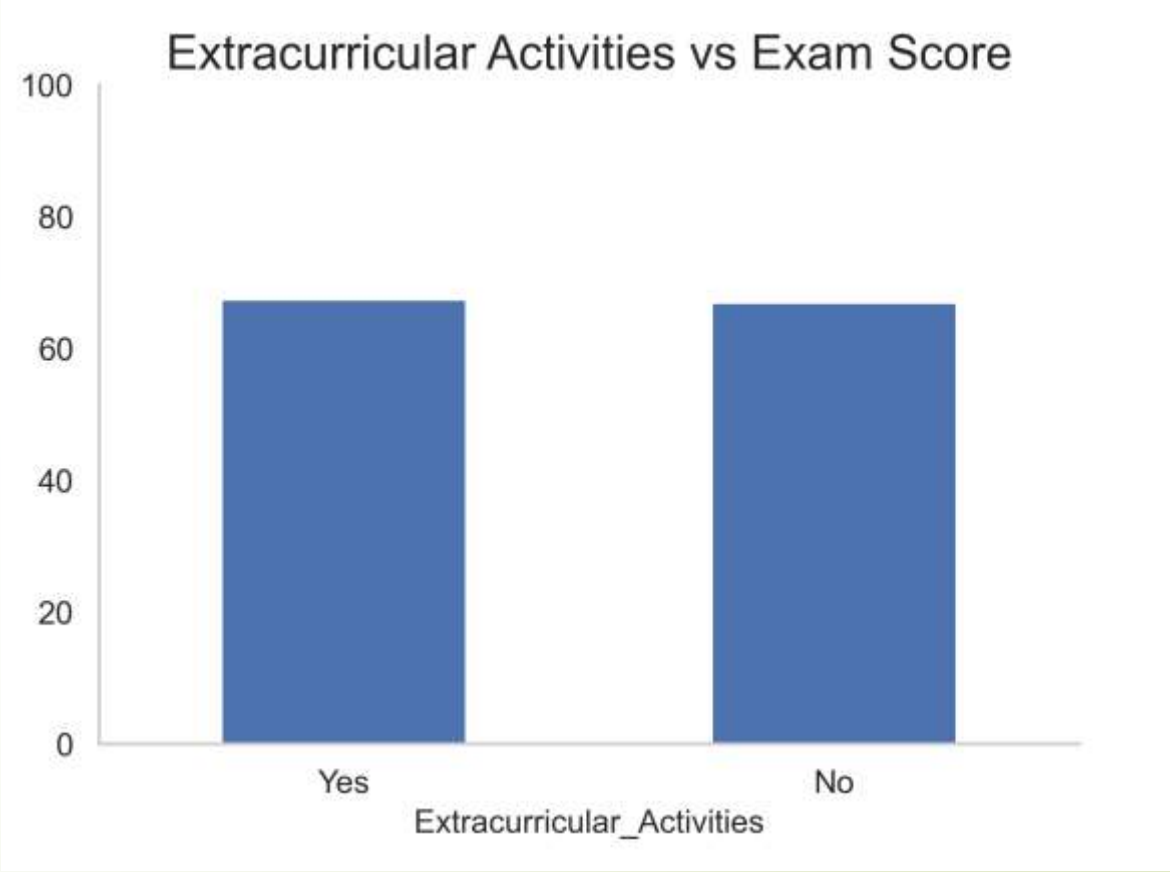

Tutoring Sessions vs Exam Score
IC=0,16
Exam Score (0-100)
100
90
80
70
60
0
1
2
3
4
5
6
7
8
Tutoring Sessions (0-8)


Sleep Hours vs Exam Score
IC=-0,017
Exam Score (0-100)
100
90
80
70
60
4
5
6
7
8
9
10
Average Sleeping Hours/Night (h/night)

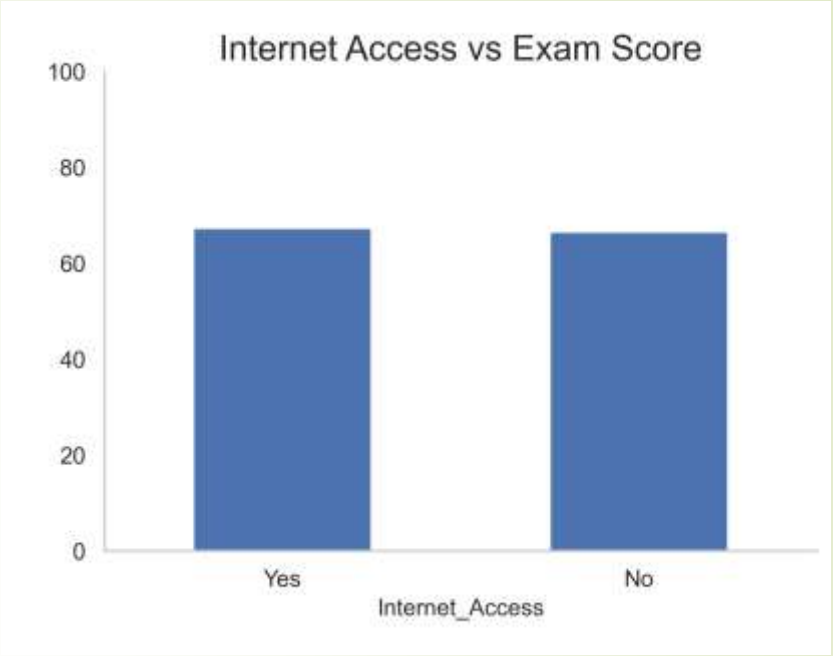

Hours_Studied
Attendance
Tutoring_Sessions
Sleep_Hours
Physical_Activity
Exam_Score
1.00 -0.01 -0.01 0.01 0.00 0.45
-0.01 1.00 0.01 -0.02 -0.02 0.58
-0.01 0.01 1.00 -0.01 0.02 0.16
0.01 -0.02 -0.01 1.00 -0.00 -0.02
0.00 -0.02 0.02 -0.00 1.00 0.03
0.45 0.58 0.16 -0.02 0.03 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

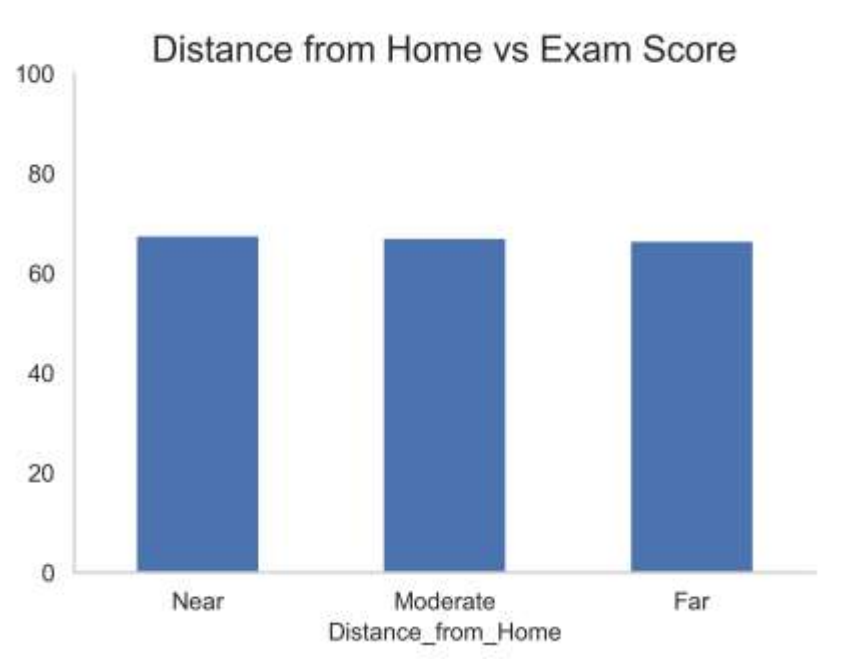
Hipótesis principal





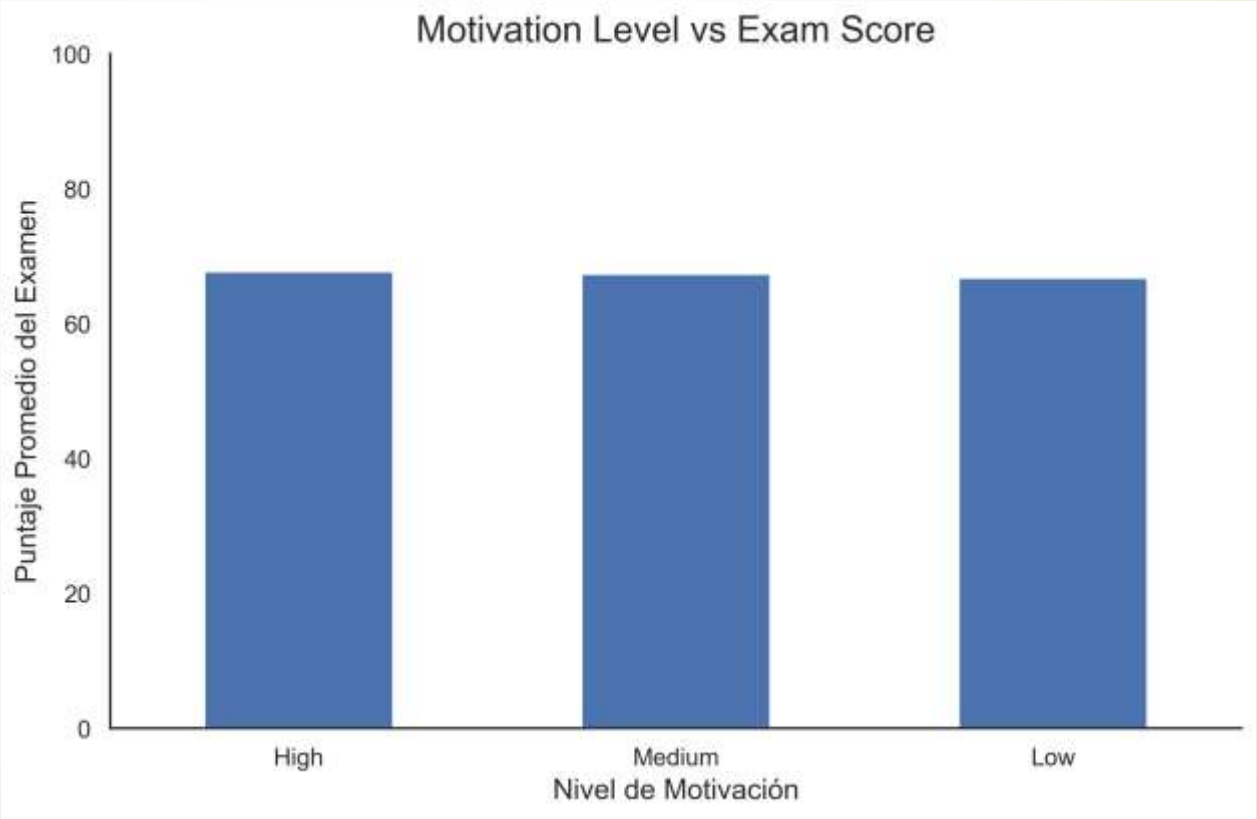
Teacher Quality vs Exam Score
100
80
60
40
20
0
High
Medium
Low
Teacher_Quality


Internet Access vs Exam Score
100
80
60
40
20
0
Yes
No
Internet_Access

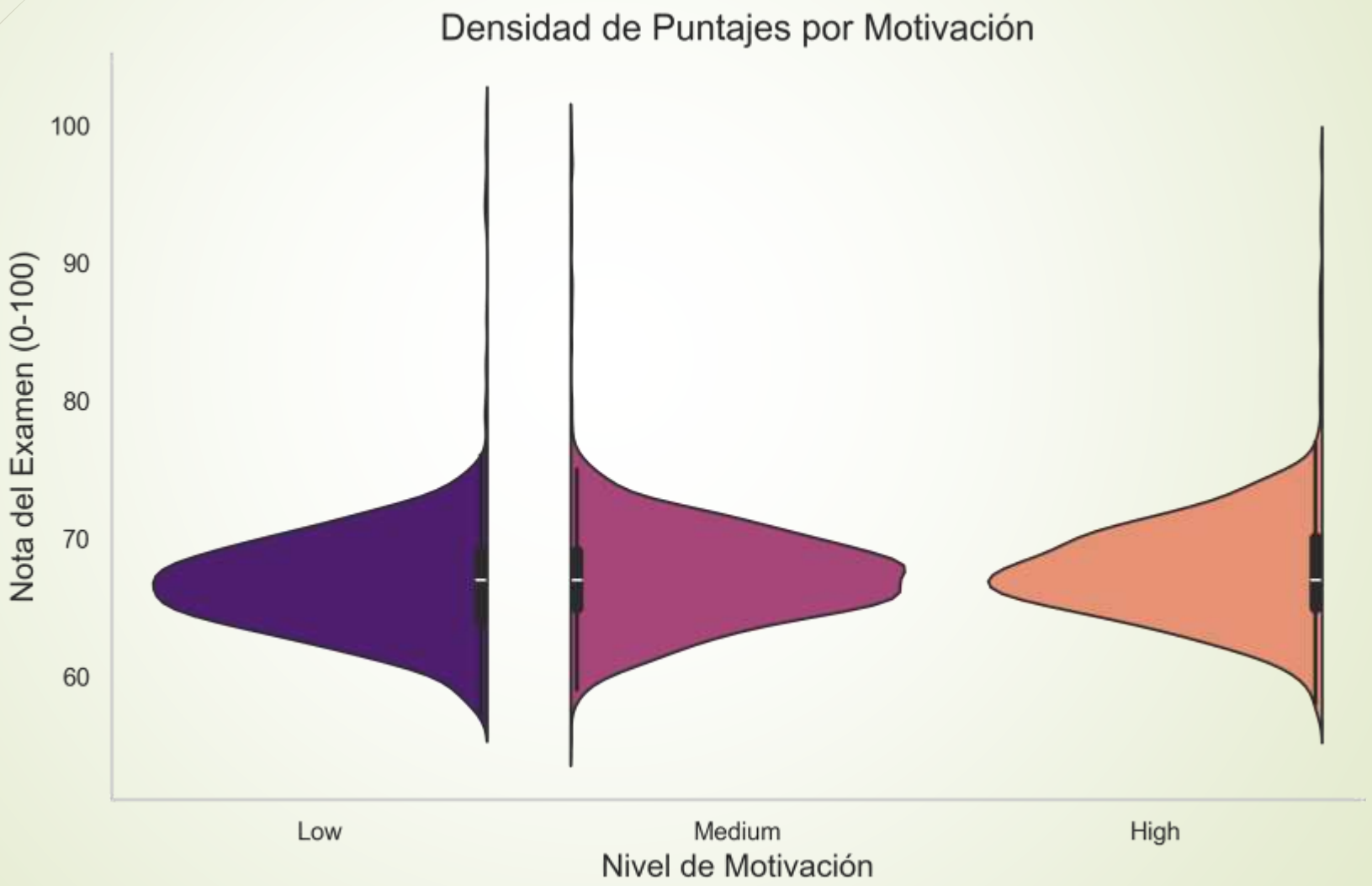

Distance from Home vs Exam Score
100
80
60
40
20
0
Near
Moderate
Far
Distance_from_Home

Subhipótesis

- "La importancia de la motivación de un estudiante a la hora de la nota final"



Subhipótesis



Conclusiones

- Hipótesis Principal:
  - "Los estudiantes con una buena organización de estudio tienen más probabilidades de sacar mejor nota"

Nivel de afectación en la nota del examen	Índice de Correlación (IC)
Attendance	0,58
Study Hours	0,46
Tutoring Sessions	0,16
Resto de variables	~0

Conclusiones

- Subhipótesis:
  - "La importancia de la motivación de un estudiante a la hora de la nota final"

Nivel de Motivación ✕ **Nota del Examen**

ESKERRIK ASKO

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